

Gerald (Jace) Curran III

Theoretical condensed matter physicist specializing in spin coherence, triplet exciton dynamics, and Monte Carlo simulation of entangled systems. Published in *PRL* and *Physical Review B*.

Education

Doctor of Philosophy in Physics

Ph.D. Defended 2025 – Awarded 2026

Lehigh University

Thesis: “The Entangled Triplet Exciton Pair in Rubrene”

Advisor: Ivan Biaggio

Master of Science in Physics

M.S. Awarded 2022

Lehigh University

Bachelor of Science in Physics (w/ Math minor)

B.S. Awarded 2020

Kutztown University

Publications

Curran, et al. *Persistence of Spin Coherence in a Crystalline Environment*

Published 2024

Physical Review Letters **133** - <https://doi.org/10.1103/PhysRevLett.133.056901>

Selected as an Editor’s Suggestion

Curran, et al. *Transport-Induced Decoherence of the Entangled Triplet Exciton Pair*

Published 2025

Physical Review B **112** - <https://doi.org/10.1103/rjsk-kjvf>

Research Experience

Graduate Research Assistant / Part-Time Postdoc

Summer 2020 – Fall 2026

Lehigh University

- Developed theoretical models of transport-induced dephasing (TID) and global decoherence explaining suppression of quantum beats in zero-field and off-axis high-field regimes in crystalline rubrene.
- Devised scheme to use experimentally determined quantum beat frequencies as a probe of zero field parameters (D and E) in acene materials, improving accuracy over previously accepted values.
- Engineered Monte Carlo Hamiltonian-swapping simulation in MATLAB, to successfully predict observables of fluorescence in acene materials, including quantum beats, TID-driven decoherence, and modulation of fluorescence by an external magnetic field.

Undergraduate Research Assistant

Spring 2019 – Summer 2020

Kutztown University

- Programmed MATLAB simulation to explore the quantum hall effect in simplified conductor models under various topologies.
- Mapped variations in the expected eigenenergy graph (the “Hofstadter Butterfly”) to changes in topology, inter-ring coupling, ring radius and perturbations.

Published Software

Beat Boost Analyzer – Companion analysis software for Picoharp .dat time-correlated photon counting files. Automates data analysis for several common observables (quantum beat frequency, amplitude, & phase; ‘boost/quench’ effect, etc), over all possible bin widths.

Boost Curve Sim – Monte Carlo simulation of the evolution of the triplet-pair spin wavefunction. Reproduces transport-induced dephasing (TID) and decoherence effects, as well as repeated fission-fusion events and spin relaxation. Required to properly model the characteristic fluorescence ‘boost/quench’ effect in rubrene.

Conference Presentations

Transport-Induced Dephasing (TID) – Short Talk	APS March Meeting 2025
Department of Energy PI Meeting – Representative for Biaggio Group	Summer 2025

Teaching Experience

Teaching Assistant <i>Lehigh University</i>	September 2020 – January 2025
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- Lead lab and recitation sections of 30 students each in introductory physics courses
- Received perfect 5/5 student evaluations, with roughly half of students responding

Adjunct Professor <i>Moravian University</i>	Spring 2026
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Adjunct Professor <i>DeSales University</i>	Fall 2026
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Work Experience

Database Administrator & Front-End Developer <i>International Institute for Learning</i>	March 2017 – May 2020
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- Constructed library of courses and course materials for international teaching organization (more than 500 courses with appendices, handouts, and other materials)
- Rescued at-risk, 1,200 question practice exam project with knowledge of python and xml

Project Manager / TARP Lead for E.PA <i>AT&T</i>	Jan 2014 – Jan 2017
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- Promoted to market TARP lead, and saved AT&T \$600,000 through tower review program
- Developed and managed dozens of automated trackers & reports, using VBA for automation